

Liangyu WU | Curriculum Vitae

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Education Background

Stanford University <i>Ph.D. Student in Physics</i>	California, U.S. 2024-Present
Shanghai Jiao Tong University <i>Bachelor of Science in Physics, School of Physics and Astronomy</i>	Shanghai, China 2020-2024
• Overall GPA: 86.1/100 (A-); Physics-related GPA: 90.1/100 (A)	
• Coursework: <i>Quantum Mechanics, Electrodynamics, Equations of Mathematical Physics, Professional Experiment, Introduction to Nuclear and Particle Physics, Methods of Experimental Nuclear and Particle Physics</i> (All graded A's)	
University of Maryland, College Park <i>Visiting International Student</i>	Maryland, U.S. Aug. 2023-Dec. 2023

Research Experience

SLAC National Accelerator Laboratory <i>Particle Physics and Astrophysics</i> <i>Research Assistant</i> (Supervisor: Dr. Julia Gonski & Prof. Dong Su)	California, U.S. Jan. 2025-Present
– Research on the ATLAS	
• On-chip machine learning for waveform analysis and data compression in dual readout calorimeters.	
• GigaBit Cable Receiver (GBCR) ASIC Testing.	
SLAC National Accelerator Laboratory <i>Particle Physics and Astrophysics</i> <i>Research Assistant</i> (Supervisor: Prof. Spencer Gessner)	California, U.S. Sep. 2024-Dec. 2024
– Research on the FACET-II	
• Compton cross sections calculation for laser-based control of beam intensity.	
• Update to the TCAV GUI and analysis tools.	
University of Maryland, College Park <i>Department of Physics</i> <i>Undergraduate Research Student</i> (Supervisor: Prof. Christopher Palmer & Prof. Sarah Eno)	Maryland, U.S. Aug. 2023-Aug. 2024
– Research on the Dual-readout Calorimetry	
• Performed Geant4 simulations of single-particle responses for diverse calorimeters.	
• Derived a formula that predicts the dual-readout-corrected energy from scintillator and Cherenkov signals alone.	
Shanghai Jiao Tong University <i>School of Physics and Astronomy</i> <i>Undergraduate Research Assistant</i> (Supervisor: Prof. Yue Meng)	Shanghai, China Oct. 2021-Feb. 2024
– R&D for a novel radon detector	
• Designed the Radic detector and determined the radon diffusion coefficient of several materials successfully.	

PandaX Collaboration

Shanghai & Sichuan, China

Undergraduate Research Student

Oct. 2021-Feb. 2023

– Work in Ultra-low background technique R&D group

- Measured radon emanation rates across diverse materials for the PandaX and JUNO experiments.
- Conducted ultra-low radioactive surface treatments of materials for TPC assembly in PandaX-4T.

– Work in PMT group

- Performed PMT high-voltage testing to ensure standard operating currents and assessed parameters.
- Participated in the installation of veto PMTs and high-voltage cables for TPC-bottom PMTs.

Tsung-Dao Lee Institute

Shanghai, China

Astronomy and Astrophysics Division

Mar. 2023-Aug. 2023

Undergraduate Research Student (Supervisor: **Prof. Masahiro Ogihara**)

– Research on the stability of unevenly spaced planetary systems

- Conducted N -body simulations to study the planetary system stability.
- Demonstrated that using evenly spaced models can overestimate orbital stability time in natural systems.

Publications

- [1] S. Eno, **L. Wu** *et al.*, On the resolution of dual readout calorimeters, <https://arxiv.org/abs/2501.15329>
- [2] S.V. Chekanov, S. Eno, S. Magill, C. Palmer, **L. Wu**, Geant4 simulations of sampling and homogeneous hadronic calorimeters with dual readout for future colliders, **NIM A**, Volume 1072, 2025, 170200, ISSN 0168-9002.
- [3] **L. Wu**, L. Si *et al.*, Design and Experimental Application of a Radon Diffusion Chamber for Determining Diffusion Coefficients in Membrane Materials, <https://arxiv.org/abs/2410.06054>
- [4] S. Yang, **L. Wu** *et al.*, The stability of unevenly spaced planetary systems, **ICARUS**, Volume 406, December 2023, 115757

Skills

Technical: C++, Python, CERN ROOT, Geant4, SolidWorks, COMSOL, LaTeX

Hardware: Ultra-low radioactive technology, High-voltage technology, PCB debugging, PMT testing

Language: English (Fluent), Mandarin (Native)